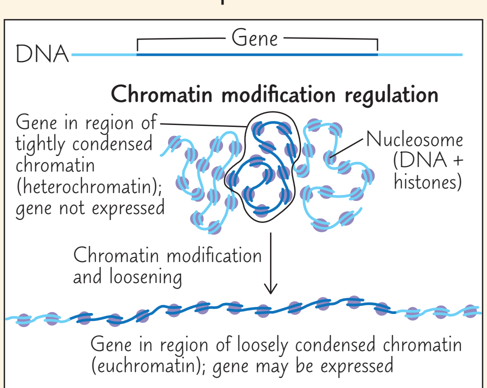
18 Regulation of Gene Expression

KEY CONCEPTS

Study Tip

Make a visual study guide: Draw a region of DNA representing a gene. For each level of gene regulation that you read about, draw a sketch of your gene being regulated at that level. A drawing of chromatin modification regulation is shown as an example.



Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 18
- Animation: Control of Gene Expression
- Animation: Causes of Cancer

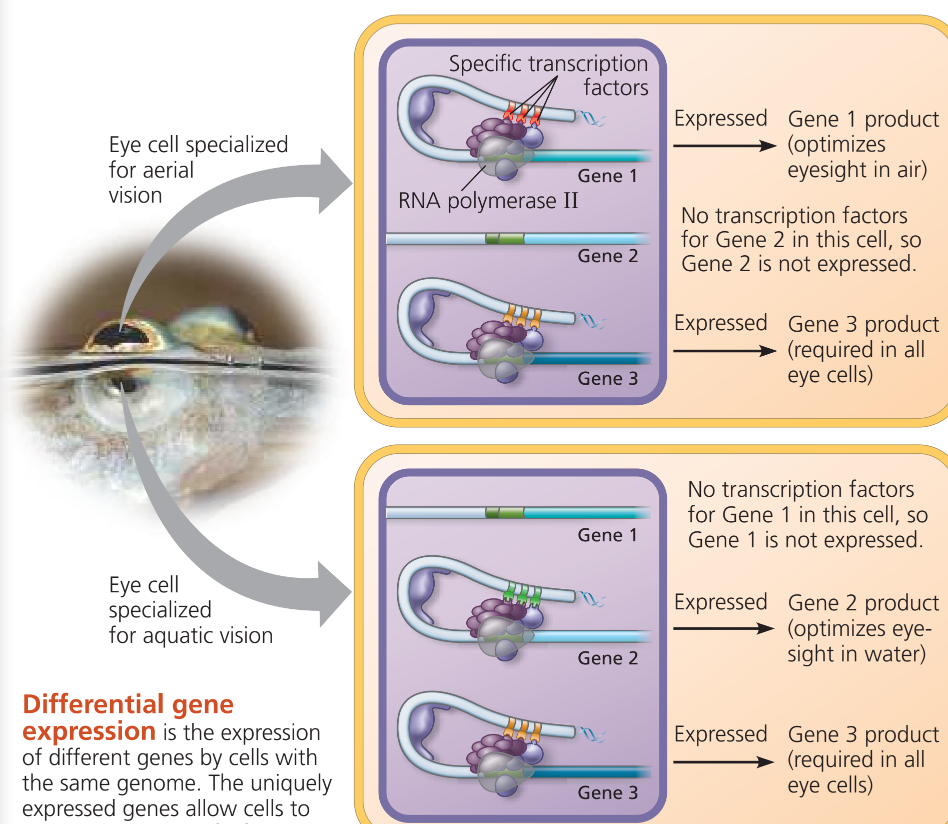
For Instructors to Assign (in Item Library)
- Tutorial: Regulation of Gene Expression in Eukaryotes
- Scientific Skills Exercise: Analyzing DNA Deletion Experiments

Ready-to-Go Teaching Module (in Instructor Resources)
- The *trp* and *lac* Operons (Concept 18.1)

Figure 18.1 *Anableps anableps*, or "cuatro ojos" ("four eyes"), glides through lakes and ponds in South America, the upper half of each eye protruding from the water. The eye's upper half is well-suited for aerial vision and the lower half for aquatic vision. All eye cells, however, contain the same genes.

How can two cells with the same set of genes function differently?

To be expressed, each gene requires a particular set of transcription factors. Different cells have different sets of specific transcription factors.



Differential gene expression is the expression of different genes by cells with the same genome. The uniquely expressed genes allow cells to carry out their specific function.